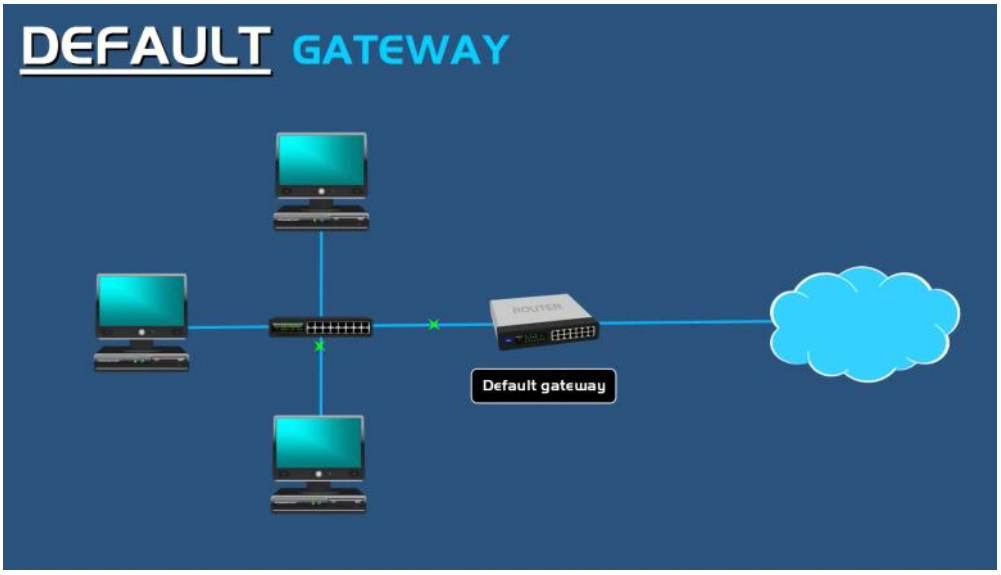
Default Gateway

Thursday, April 23, 2026 5:57 PM

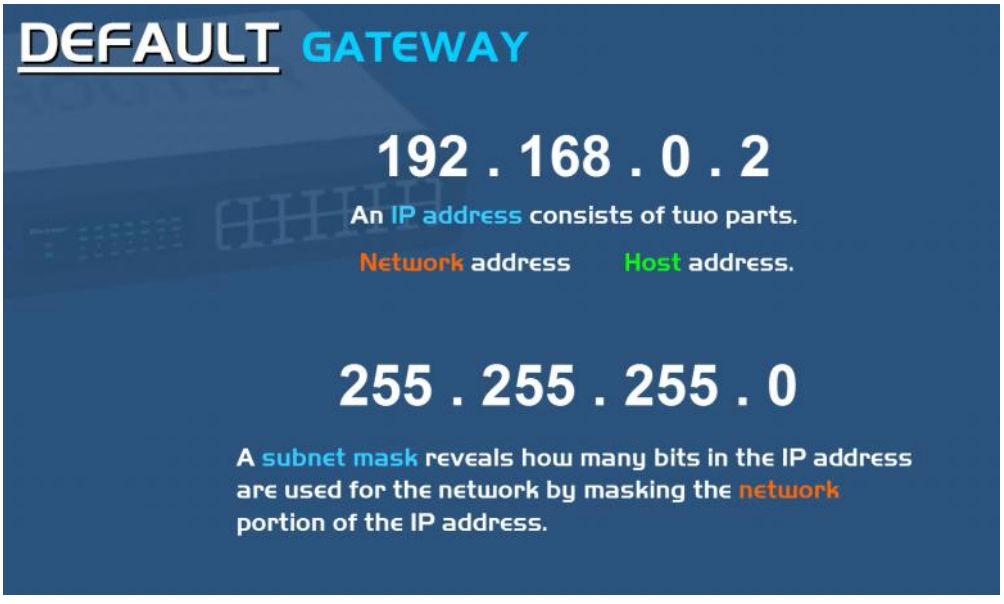
Default Gateway
A default gateway is a network device, typically a router, that acts as a doorway to forward data between different networks.

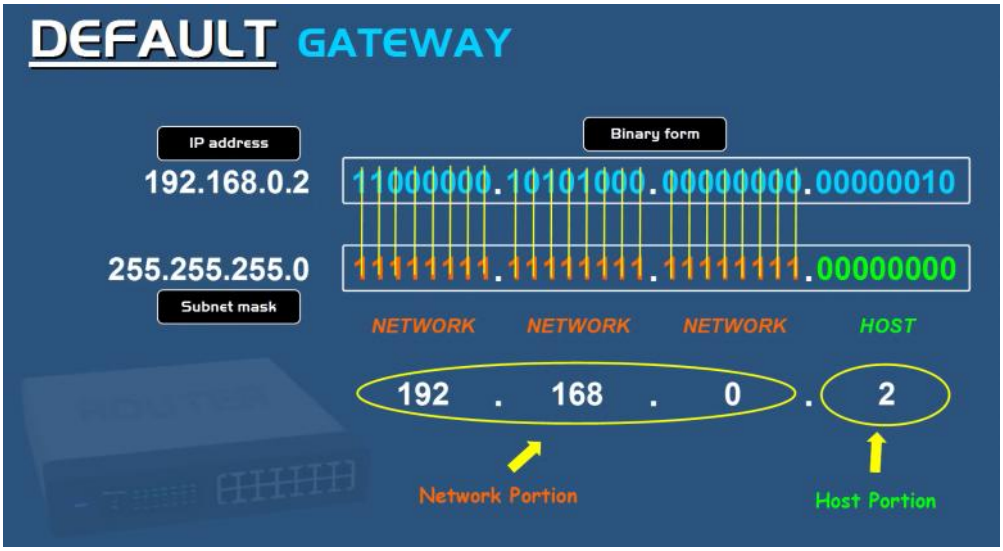


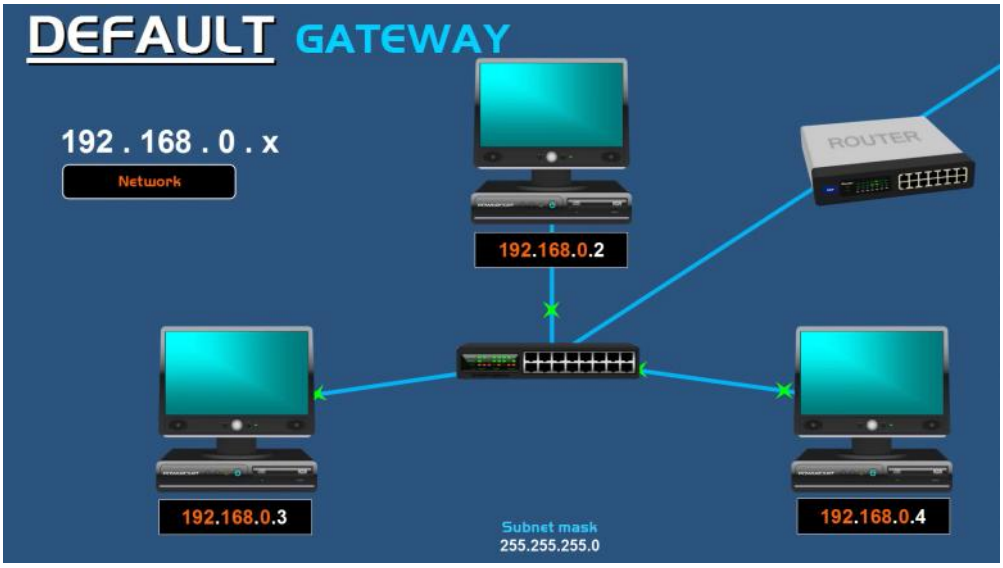
How Communication Works:

- Local Communication: Devices on the same network communicate directly through a switch. They do not need the default gateway because they share the same network address portion.
- Remote Communication: When a device needs to send data to a different network, it uses the default gateway to bridge the gap.

Determining Network Destinations:
Devices use their IP address and subnet mask to identify if a destination is local or remote:



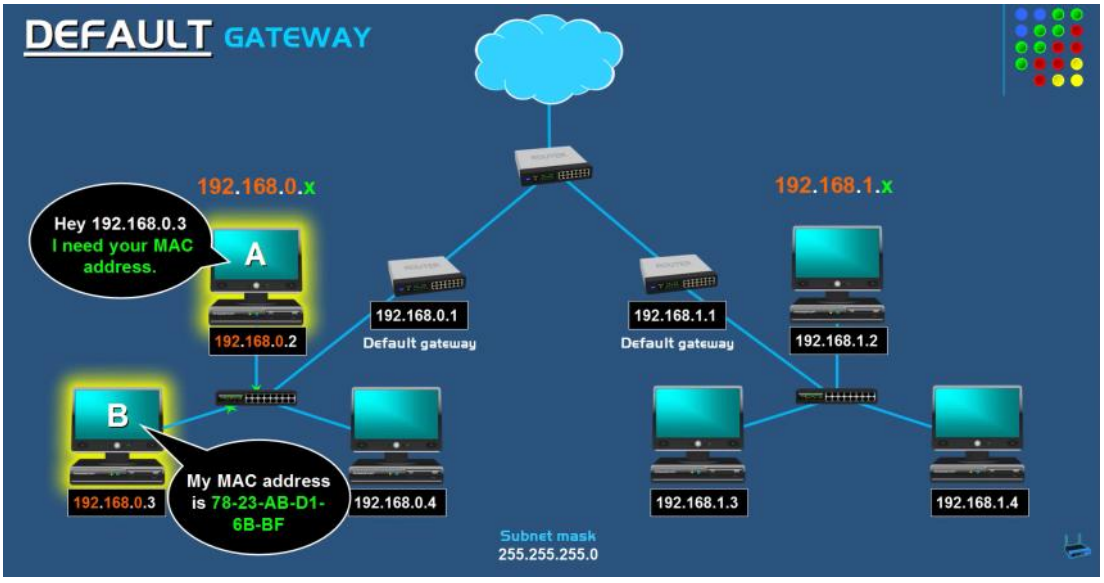


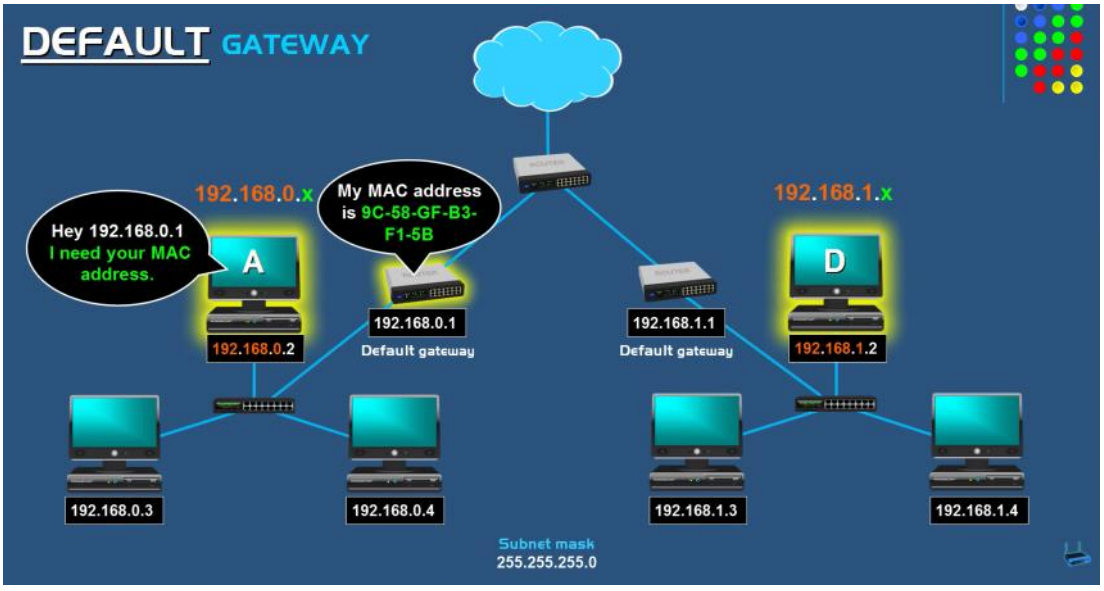


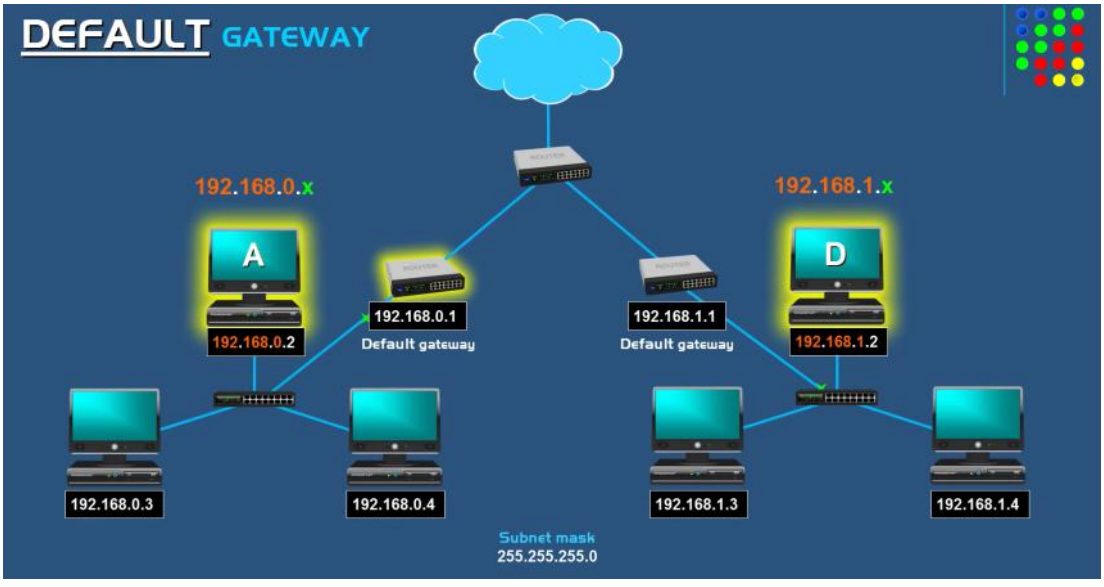
- The subnet mask indicates which part of the IP address represents the network and which part represents the host.
- If the network portions of two IP addresses match, they are on the same network.
- If the network portions are different, the source device recognizes that the destination is on another network and sends the data to its default gateway.

MAC Addresses and ARP:

To finalize communication, devices use ARP (Address Resolution Protocol) broadcasts to find the MAC address of their target. If the destination is on a different network, the source sends the ARP request to the default gateway's MAC address instead of the remote device, as ARP broadcasts cannot pass through a router.


DEFAULT GATEWAY
192.168.0.x
192.168.1.x
Hey 192.168.0.3
I need your MAC address.
A
192.168.0.2
192.168.0.1
Default gateway
192.168.1.1
Default gateway
192.168.1.2
B
192.168.0.3
My MAC address is 78-23-AB-D1-6B-BF
192.168.0.4
192.168.1.3
192.168.1.4
Subnet mask
255.255.255.0


DEFAULT GATEWAY
192.168.0.x
192.168.1.x
Hey 192.168.0.1
I need your MAC address.
My MAC address is 9C-58-GF-B3-F1-5B
A
192.168.0.2
192.168.0.1
Default gateway
192.168.1.1
Default gateway
D
192.168.1.2
192.168.0.3
192.168.0.4
192.168.1.3
192.168.1.4
Subnet mask
255.255.255.0


DEFAULT GATEWAY
192.168.0.x
192.168.1.x
A
192.168.0.2
192.168.0.1
Default gateway
192.168.1.1
Default gateway
D
192.168.1.2
192.168.0.3
192.168.0.4
192.168.1.3
192.168.1.4
Subnet mask
255.255.255.0